Acids and Bases in Organic Chemistry

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Some problems and ideas to guide of class discussion of acid equilibrium[1]

[1] Read ch. 5.1 to 5.4 of the textbook.

Complete Before the Class Meeting

Here are some questions to remind you of what you learned about acids and bases in your previous chemistry classes.

1. Write our the chemical equation for the acid equilibrium of acetic acid. the write out the equation for the acid dissociation equilibrium constant, K_a.

2. The pK_a value for acetic acid is 4.75. What is the value of the equilibrium constant for acid dissociation, K_a

3. Tell me how you would mix up a 100 mM acetate buffer so that when you measure the pH it will be at or very near a pH value of 5.1.

4. *para*-Nitrophenol is a useful leaving group for measuring rates of ester hydrolysis. It is a yellow colour when in its deprotonated form. I am performing an enzymatic hydrolysis of *p*-nitrophenylacetate as a probe for ester hydrolysis by a suite of hydrolases that are used in industry. I am doing the reaction at a pH value of 7.5. What percent of the nitrophenol is it the yellow coloured form? Explain why the pK_a value of *p*-nitrophenol is lower than the pK_a value for unsubstituted phenol.

5. Saccharine is (or used to be) present in soft drinks as its sodium salt. Can you show me the acidic proton of saccharine and propose a reason why it is never likely to be found in your coffee sweetener as anything other than a salt? Back up your answer with facts.

6. Write our the famous Hendersson-Hasselbach equation and show me how it is just an alternate version of the acid equilibrium expression.

Table 1: Selected pK_a values.

Acid	pK_a
Acetic acid	4.75
Phenol	9.95
p-Nitrophenol	7.10
Saccharine	1.60
Aniline	4.62
4-nitroaniline	1.11

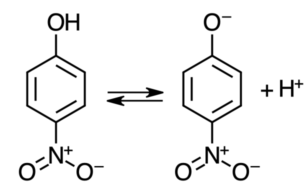


Figure 1: Acid equilibrium for nitophenol

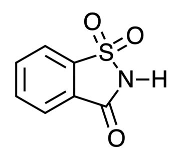


Figure 2: The structure of saccharine

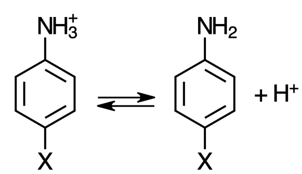


Figure 3: Acid equilibrium for anilines

Textbook Questions

Try these problems: Ch. 5: 1–6, 9, 11–13 & 19–21

Questions for the Class Meeting

1. What is the *p*H of 0.01% (w/v) HNO_3 in water? What is the *p*H of 85% (w/v) HNO_3 in water?[2] Use the data below to answer this question.

[2] The molar mass of 100% HNO_3 is 63.01 g/mole and the density is 1.51 g/mL

	% in acidic form			
% wt HNO_3	4-nitro-aniline	4-chloro-2-nitroaniline	4,6-dichloro-2-nitroaniline	6-bromo-2,4-dinitroaniline
0.1	14.53	0.15	0.00	0.00
4	92.80	10.30	0.03	0.00
40	99.96	95.63	5.32	0.01
85	100.00	99.99	95.72	2.05
100	100.00	100.00	99.80	31.37

Table 2: Observed ionization of selected indicator acids in mixtures of HNO_3. See table 1 for pK_a value

2. What is an acidity function? Compare the pH and H_0 acidity functions. Are they really different?

Explain the relative pK_a values in the following series.

3. The pK_a value for several series of hydrides are presented in tables 3, 4, 5 and 6. Explain the trend in each series.

Table 3: pK_a values of 2[nd] row hydrides.

CH_4	NH_3	H_2O	HF
48	33	16	3

Table 4: pK_a values of halogen hydrides.

HF	HCl	HBr	HI
3	-6	-9	-10

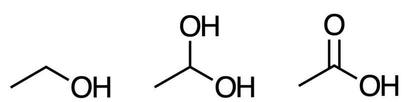
Table 5: pK_a values of oxygen hydrides.

HClO	$HClO_2$	$HClO_3$	$HClO_4$
7.5	-6	-9	-10

Table 6: pK_a values of carbon hydrides.

HC≡CH	$H_2C{=}CH_2$	$H_3C{-}CH_3$
50	44	25

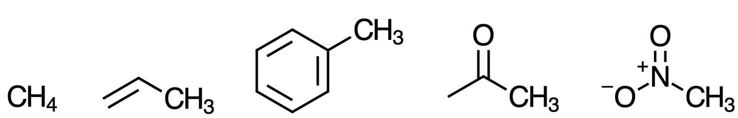
4. Explain the trend in table 7 below.



16	14	5

Table 7: pK_a values of hydroxides.

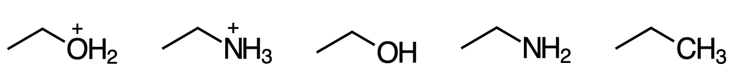
5. Explain the trend in table 8 below.



48	42	38	20	10

Table 8: pK_a values of carbon hydrides.

6. Explain the trend in table 9 below.



-2	9	16	35	50

Table 9: pK_a values of acids.

Acidity Function Data

The protonating power (*p*H or H_0) of acid/water mixtures[3] has been determined using substituted anilines[4] and other probes.[5] See figure 4 for some common mixtures.

[3] See ch. 5.2.5

[4] For a useful review see "Acidity functions: an update." R.A. Cox, K. Yates, *Can. J. Chem.*, **1983**, *61*, 2225–2243. https://doi.org/10.1139/v83-388

[5] For examples of other acidity function scales see "Protonation and Solvation in Strong Aqueous Acids." E.M. Arnett, G. Scorranno, *Adv. Phys. Org. Chem.*, **1976**, *13*, 83–153. https://doi.org/10.1016/S0065-3160(08)60213-0

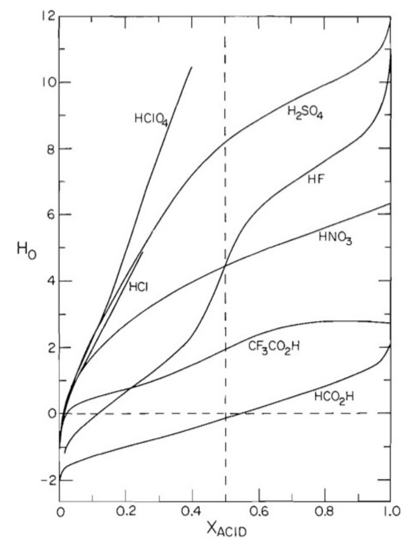


Figure 4: H_0 values vs. mole fraction for selected mixtures of acids and water.[4]

Brønsted Acid and Base Catalysis

Many organic reactions that we have reviewed already required protonation or deprotonation as part of the mechanism. They would not occur at a measurable rate without strong acid or base present. We will discuss this topic in more detail later but let us make some definitions now.

Acid and base catalysis is defined according to how it is involved in the rate-determining step and therefore how it affects the rate law.

Specific catalysis: the proton transfer takes place before the RDS as part of a acid/base equilibrium reaction. The portion or protonated intermediate will depend only on the acidity function (*p*H) of the medium.

General catalysis: the proton transfer takes place during the RDS. There is no acid/base equilibrium previous to that step. The rate will depend on the amount of acid/base present, whether it is as solvent (e.g. H_3O^+ or HO^-) or as buffer.[6]

[6] See ch. 9.3 for more about acid and base catalysis.

Examples of Acid and Base Catalysis

E1 elimination of alcohols in strongly acidic conditions proceeds by creating the protonated alcohol intermediate, which then undergoes elimination with the rate-determining step being carbocation formation. The alcohol group is just as easily protonated by strong acids as water (the pK_a of the protonated alcohol is approximately −1.) So there will be a significant population of the protonated intermediate. This is an example of specific acid catalysis.

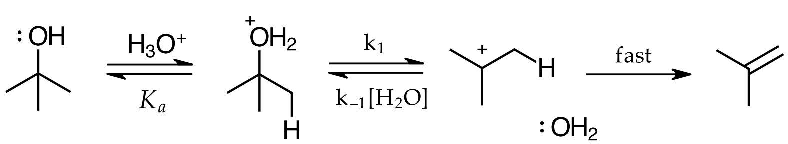


Figure 5: Acid-catalyzed dehydration of t-butanol

E2 elimination is strongly basic conditions does not proceed via a deprotonated intermediate. The pK_a of an alkyl group is near 50. It is almost impossible to remove the proton using a base in this case. The base will begin connecting with the proton as the leaving group departs. We never create a high energy carbanion or carbocation intermediate. The leaving group departure is assisted by the base during the RDS. This is an example of general base catalysis.

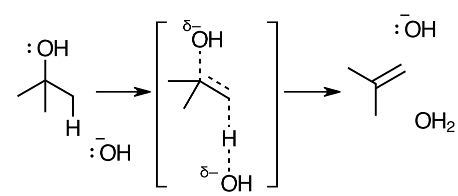


Figure 6: Base-catalyzed dehydration of t-butanol

A Look Ahead to Kinetics

The solvolysis of t-butanol is a classic example of the S_N1 reaction mechanism. In neat acetic acid we will observe the appearance of t-butylacetate. The rate of reaction depends on the acidity of the reaction mixture. We will now answer the following questions.

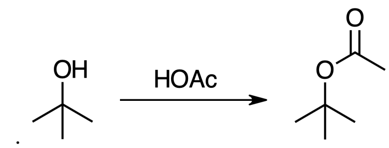


Figure 7: fig:Solvolysis of t-butanol in acetic acid

1. Write out a scheme for the stepwise mechanism of this reaction
2. Derive the rate law for this reaction. Then modify the rate law so that the concentration (activity) of acid is expressed in terms of an acidity function such as pH or H_0.
3. Examine the rate law and determine if it describes a first-order reaction. Can it be rewritten in a form that describes a first-order reaction? If so, write it out as a first order rate law.
4. Examine the law and determine the change in rate if the acidity is increased by a factor of 3. How might one increase the acidity of an acetic acid mixture?[7]

[7] "The Acidity Scale in Glacial Acetic Acid. I. Sulfuric Acid Solutions. $-6 < H_0 < 0$." N.F. Hall, W.F. Spengeman *J. Am. Chem. Soc.*, **1940**, *62*, 2487-2492 https://doi.org/10.1021/ja01866a062

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will...

- have reviewed more organic chemistry mechanisms and are able to describe reactions that feature Brønsted acid and base catalysis.
- have recalled how to calculate acid equilibria and ionizations using the acid dissociation equilibrium constant and the acidity function of the medium.
- have expanded their definitions of acidity functions beyond pH and become familiar with the Hammett acidity function, H_0
- have gained an understanding of factors that affect the acidity of a group in a molecule.
- **Computer Skill**: be able to use an interactive *Python* notebook to perform and document calculations.